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To Whom It May Concern,

I am pleased to write this letter of recommendation for **Mr. Sarthak Fulzele**, who has been an exemplary student in the Department of Mechanical Engineering. I have had the privilege of teaching and supervising Sarthak during his undergraduate studies, and I can confidently attest to his exceptional academic abilities, strong work ethic, and dedication to research. Sarthak has consistently demonstrated outstanding performance in my classes. His analytical skills and understanding of complex mechanical engineering concepts are truly remarkable. He has shown a keen interest in expanding his knowledge and has actively participated in classroom discussions, often providing insightful contributions that enhance the learning experience for his peers.

In August 2024, Sarthak embarked on his bachelor's thesis project under my supervision on the topic of "Ammonia and its blending with micro-mix of Hydrogen and Hydrogen Peroxide." Despite the challenging nature of this research, Sarthak has made significant progress in a very short time. Within just one month, he has conducted thorough literature reviews, designed and executed numerical simulations, and analysed data with precision and accuracy. His project is particularly noteworthy for its innovative use of an artificial neural network model to predict combustion characteristics and optimize the efficiency of hydrogen-ammonia fuel mixtures. His diligence and commitment have led to substantial findings, and we are now in the final stages of preparing a conference as well as a journal paper for submission.

Sarthak's ability to manage his time effectively and his dedication to his research are commendable. He has demonstrated strong problem-solving skills and an innovative approach to tackling complex issues. His work on the Hydrogen-Ammonia Combustion project, including the integration of artificial intelligence, is a testament to his ability to conduct high-quality research and contribute valuable insights to the field of mechanical engineering. Moreover, Sarthak possesses excellent communication skills, both written and verbal. His ability to articulate his ideas clearly and persuasively has been evident in his project reports and presentations. He has also shown great teamwork and collaboration skills, often working effectively with his peers and contributing positively to group projects. During his research internship, he showed consistent results with thorough and concise progress reports. He was in constant touch with his mentors throughout the week, rectifying the problems discussed. Mr. Sarthak was very consistent in his efforts, providing weekly updates and demonstrating regular progress.

In addition to his academic and research accomplishments, Sarthak is a person of high integrity and strong character. He approaches his work with a sense of responsibility and professionalism that is truly commendable. I have no doubt that Sarthak Fulzele will continue to excel in his future endeavours. He has the potential to make significant contributions in any field he chooses, and I wholeheartedly recommend him for any position or program he applies to. The undersigned may be contacted for further information about the student. I wish Mr. Sarthak all the success in his academic career.

Yours sincerely,

Dr. V. Mahendra Reddy